



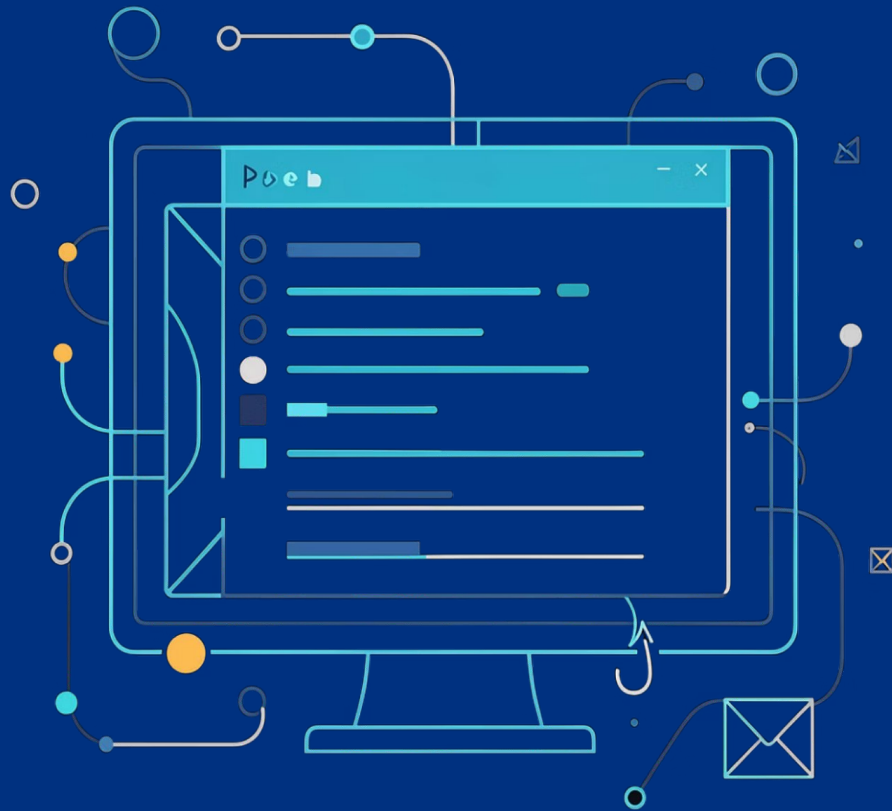
PhishGuard AI

AI-Based Phishing Website Detection System using Machine Learning

CRYPTOGRAPHY & NETWORK SECURITY

WEB-BASED SECURITY APPLICATION

The Growing Phishing Threat



Common Cyber Threat

Phishing attacks are one of the most prevalent cyber threats today, with attackers creating fake websites to steal sensitive information.

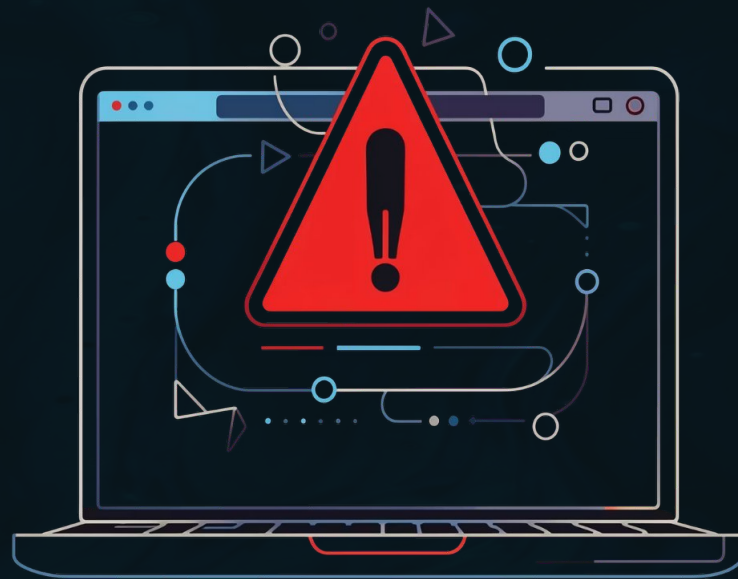
User Vulnerability

Users often cannot identify malicious URLs easily and get tricked into entering passwords and banking details.

The Gap

There is a critical need for an intelligent, real-time system to detect phishing websites using AI and security analysis.

Problem Statement



Rising Fake Websites

The increasing number of fake and phishing websites makes it harder for users to distinguish legitimate sites.

Browser Limitations

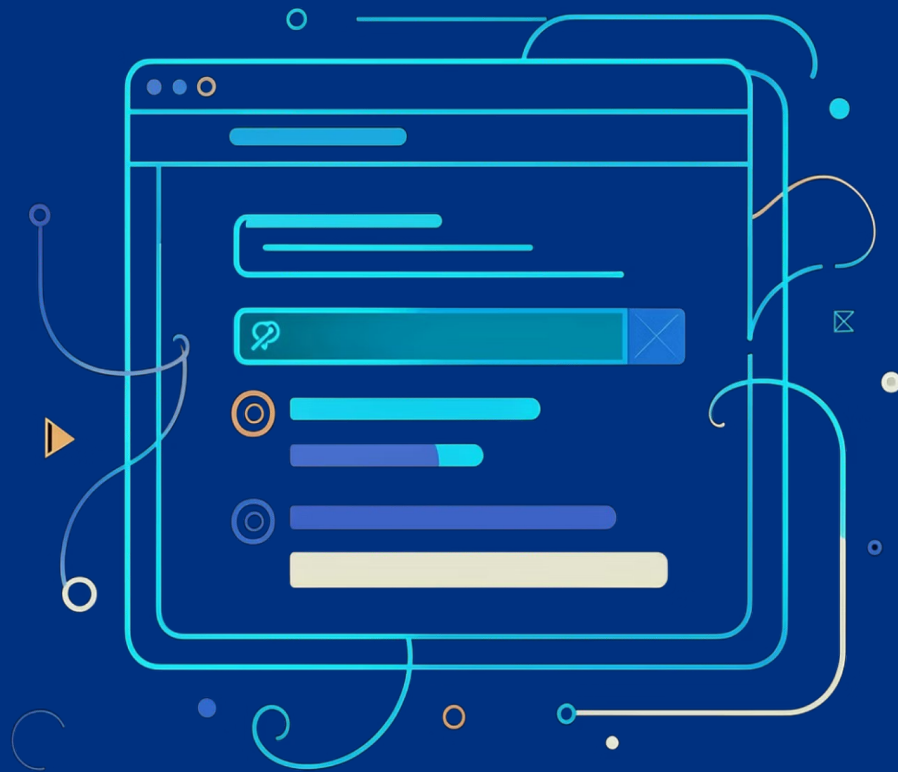
Traditional browsers cannot detect all phishing sites, leaving users exposed to credential theft.

Detection Gap

There is a lack of real-time intelligent detection systems — a fast, accurate, and user-friendly phishing detection tool is urgently needed.

Proposed Solution: PhishGuard AI

A web-based AI system where users enter a URL for scanning. The system analyzes URL structure, SSL certificate, and domain age, then applies machine learning prediction to output a result: **Safe / Suspicious / Phishing** with a risk score.



1

URL Input

User enters a URL for scanning

2

Multi-Layer Analysis

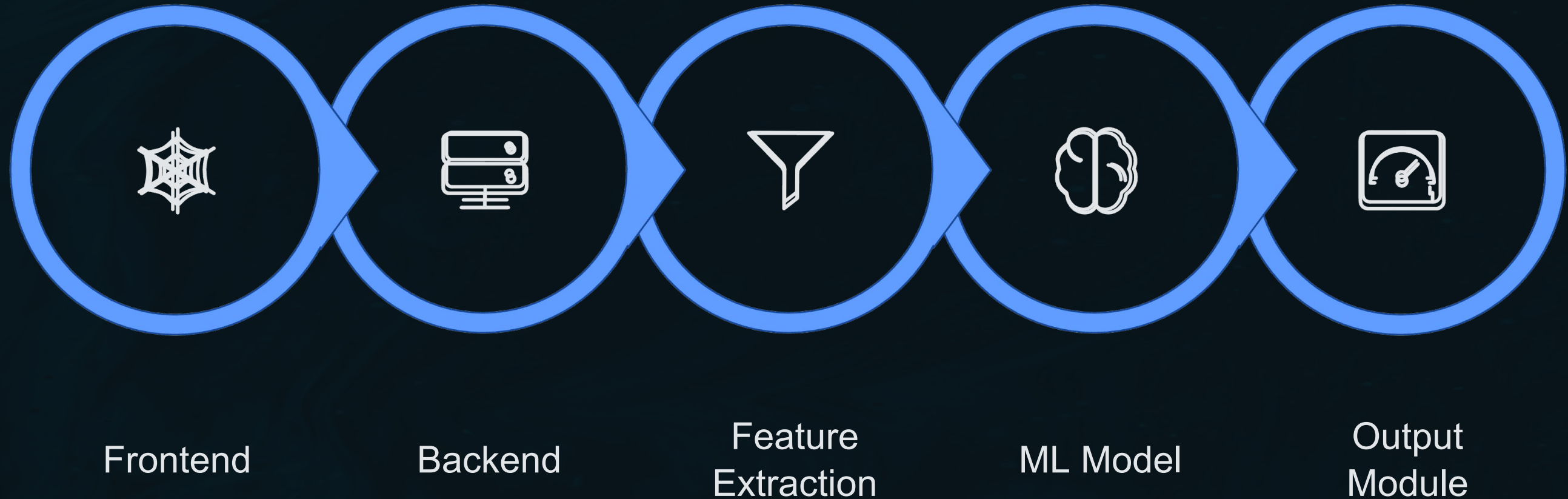
URL structure, SSL certificate, domain age, and ML prediction

3

Risk Score Output

Safe / Suspicious / Phishing with risk score

System Architecture



The system processes each URL through five sequential layers — from user input on the frontend through Flask backend processing, feature extraction, ML-based phishing prediction, and finally a result dashboard displaying the classification and risk score.

How It Works

01

URL Submission

User submits a URL through the web interface

02

Feature Extraction

System extracts security features from the URL

03

SSL & Domain Verification

SSL certificate and domain age are verified

04

ML Analysis

ML model analyzes extracted features

05

Classification

Classifier predicts: Safe, Suspicious, or Phishing

06

Result Display

Result displayed with risk score and explanation

Machine Learning Model

Features Used

URL Length

Special
Characters

HTTPS Presence

Domain Age

Subdomain Count

Algorithms & Dataset

Trained on a phishing and legitimate URLs dataset using **Random Forest** and **Decision Tree** classifiers. Output includes classification with a confidence score.

Technology Stack



Frontend

HTML, CSS, JavaScript, Bootstrap



Backend

Python Flask



Machine Learning

Scikit-learn, Pandas, NumPy



Database

SQLite



APIs & Libraries

WHOIS lookup, SSL verification
(Python SSL module)



Features & Advantages



Real-Time Detection

Instant phishing detection with an easy-to-use web interface



AI-Powered

Intelligent ML-based prediction with risk score and explanations



Security Validation

SSL and domain security validation to prevent data theft and cyber fraud



Lightweight & Scalable

A lightweight and scalable system architecture

Conclusion & Future Scope

Conclusion

PhishGuard AI successfully detects phishing websites using AI and security analysis, helping users identify threats before accessing harmful websites.

Future Scope

- **Browser Extension**
Browser extension integration for real-time protection
- **Email Phishing Detection**
Extend detection to email-based phishing attacks
- **Mobile App**
Mobile app version for on-the-go scanning
- **Blockchain Verification**
Blockchain-based URL verification
- **Deep Learning Upgrade**
Advanced deep learning model upgrade